RDIMS 596918

QUALIFICATION STANDARD AND PLAN (QSP)

Maritime Surface and Sub-Surface (MARS) (00207),
Naval Combat Information Operator (NCIOP) (00114),and
Sonar Operator (SONAR OP) (00324)

Submarine Warfare Director (SMWD)

(NQual – AILT)

MITE Course IDs:
Residential Phase – 121009
OJPR - NA

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through its Training Authority (TA), Naval Personnel and Training Group (NPTG); the Commander Canadian Submaine Force (CCSF), and Commandant of Naval Fleet School Atlantic (NFS(A). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

REVISION	APPROVAL DATE	AUTHORITY
1.0	30 Jul 04	DMTE
2.0	2015	CNPTG – Ch 1&2 Cmdt CFNOS – Ch 3&4
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FOREWORD

1. This training document is developed under the authority of the Commander of the Royal Canadian Navy (C RCN), delegated to the Commander Naval Personnel and Training Group (CNPTG), and in accordance with the following references:

   a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-001 Occupation Specification (Maritime Surface and Sub Surface) dated 14 Dec 12;

   c. A-PD-055-002/PP-002 Occupation Specification (SONAR Op) dated 18 Dec 12;

   d. A-PD-055-002/PP-002 Occupation Specification (NCI Op) dated 25 Jun 14;

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA for approval.

5. The designated TEs for this training is indicated below:

TE	Phase / Module	MITE Course ID	NQual
CFNOS	All	121009	AILT

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER CANADIAN SUBMARNINE FORCE

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Warfare Director aboard HMC Submarines. This QSP is designed to assist training establishments in developing, conducting, evaluating and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22- Qualification Standard and Plan, para 4.5. For this course, a period of instruction is 55 minutes in duration. For this course, a period of instruction is 50 minutes in duration.

	Periods
Actual Training Time (ATT) (Sum of all EOs/POs excluding PC&ECs)	244
Enabling Checks (ECs)	55.0
Performance Checks (PCs)	60.0
Sub-total	359 45 **trg days**
Allocation of BAT, TAT, TTT, PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	8 (1 day)
Training Administrative Time (TAT)	18
Training Travel Time (TTT)	0
Physical Fitness (PT)	27
Sub-total	412.0 **52 trg days**
Course Duration Calculation (ATT+PCs&ECs+BAT+TTT+TAT+PT) **8 (50 min) periods** * **Partial days must be rounded up**	412 **52 trg days**

RESOURCES

3. Identify key personnel and facilities required to conduct the training. Additional training establishment resources can be listed in Annex H if required.

Required Course Staffing				
Position	Rank	Qualifications Required	**QTY**	**PO / EO**
Course Officer	Lt(N)	• SMORO (AILU); • Instructor Supervisor (AHCJ) (Preferred)	1	EO 002.01, PO 002

Required Course Staffing				
Position	Rank	Qualifications Required	**QTY**	**PO / EO**
Senior Instructor	Lt(N)	• SMWD (AILT); • Instructional Techniques (AHCH); • Alternate Instructional Techniques (AIMU) and/or Instructor Supervisor (AHCJ) (Preferred)	1	PO 001, EO 003.01, PO 003, EO 004.01, EO 004.02, PO 004, EO 005.01, EO 005.02, PO 005
Sonar/ Oceanography Instructor	MS	Sub Sonar UC1 (AILE) - At sea experience as UC1 required	1	EO 001.01, EO 001.02
Communications Instructor	PO2	Sub Nav Comm (AIIW) - At-sea experience as RS required	1	EO 005.01
CCS 876 Instructor	PO2	Sub CRWK(D) (AIMZ) - At sea experience required	1	EO 005.01
Intelligence Officer	Lt(N)	Employed at TRINITY	1	EO 005.02
AJAG Officer	Lt(N)	ROE	1	EO 005.02
Mine Warfare Officer	Lt(N)	Employed at CFMWC	1	EO 005.02

CRITICAL TRAINING RESOURCES	**NO OF DAYS**
SCTT	25.5
Secret Laptops (1 per trainee)	71
SDM Laptops (1 per trainee)	71
Classroom with DWAN access	71

COURSE CAPACITY

1) Planning for the conduct of training is based on the guidelines in NPTORD 4500-40 – Quantity Control Process for the NTS:

   a. Maximum – 8;

   b. Minimum – 4; and

   c. Optimum – 6.

   Note: When determining Max, Min and Optimum capacity, the QSPWB is to consult with the TE(S) regarding number of students per course which is based

on resource limitations (money, time, instructors to teach and supervise)), equipment limitations (boats, tools, workstations), classroom limitations (desks and chairs), safety concerns and instructional techniques (syndicate size). Operational considerations can be captured if required under Optimum Capacity.

MITE

5. Loading. The authority to load members onto this course lies with the Training Establishment (TE). NPTG is to be included as an INFO addressee on all loading messages.

6. Scheduling. Each session of this course shall be scheduled IAW NPTORD 4500-1 - Individual Training and Education Policy and NPTORD 4500-40 Quantity Control Management for the Naval Training System; in MITE by the Training Establishment conducting the training, and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

7. Prerequisites. To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

d. Basic Submaine Course course (NQual AILS, MITE Crse IDs 109201,109285); or

e. 2OOW (MITE Crse IDs 120231/1120232);

f. Have CO's recommendation; and

g. Possess a valid Level II Security Clearance.

8. Qualification. Successful completion of this course constitutes the primary means of attaining the Submarine Warfare Director qualification (AILT). Additionally, the following MITE Course IDs will be granted:

a. Submarine Warfare Director (AILT) - MITE Crse ID121009

COURSE REPORTING

9. Upon passing this course, trainees shall receive a Course Report (CF 377) and/or Certificate of Military Achievement (CF 289) completed IAW CMP Instruction 01/06 - Course Reporting and Certificates and NPTORD 4500-51 - Course Reports and Certification.

EVALUATION

10. Program evaluation shall be conducted IAW the guidance provided in A-P9-050-000 CFITES Vol. 11 - Evaluation and Validation Techniques.

VALIDATION

11. This training will be validated every five years CFITES Vol. 8 - Validation of Instructional Programs and based on the task list in Annex B.

PRIOR LEARNING ASSESSMENT

12. PLARs shall be conducted in accordance with NPTORD 4500-27 – PLAR, DAOD 5031-1 - Canadian Forces Military Equivalencies Program and CFITES Vol 12 - Canadian Forces Military Equivalencies Programme: Prior Learning Assessment.

SPECIAL INSTRUCTIONS

13. Individuals successfully completing this course will also be considered Submarine Weapons Certified IAW CFCD 122.

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
CHAPTER 4: COMMANDER CANADIAN SUBMARNINE FORCE

PO 001 Manage Sensor Information Flow

1. Performance. Manage Sensor Information Flow

2. Conditions

   a. Given:

      (1) Submarine;

      (2) Supervision; and

      (3) References.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) At Sea; and

      (2) Watch Dived.

3. Standard. The SMWD shall, IAW references, (T0010) manage sensor information flow by:

   a. Interpreting the effects of the environment on submarine sensors;

   b. Applying principles of sonar search procedures; and

   c. (T0006) Provide advice to command on the use of sensors.

4. References

A8, A9, A15, A16, A22, A24, A27, B3, B4, B5, B7, B14, B16, B19, B23, D5

5. Task Numbers

T0006, T0010

6. Training Limitations

Training will be conducted in SCTT.

PO 002 Operate Submarine Launched Weapons

1. Performance. Operate Submarine launched weapons

2. Conditions

   a. Given:

      (1) Submarine;

      (2) Weapon Acoustic Preset Program (WAPP);

      (3) Watch teams;

      (4) References; and

      (5) Supervision.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) At Sea; and

      (2) Watch Dived.

3. Standard. The SMWD shall, IAW references, (T0005) operate submarine launched weapons and weapon control systems, to include:

   a. Conducting pre-firing activities;

   b. Firing the weapon; and

   c. Conducting post firing activities.

4. References

A1, A2, A3, A8, A9, A10, A11, A12, A15, A23, B3, B4, B6, B21, B22, D5, D7, D8

5. Task Numbers

T0005

6. Training Limitations.

Training will be conducted in SCTT.

PO 003 Conduct Eyes Only Attacks

1. Performance. Conduct Eyes Only Attacks

2. Conditions

   a. Given:

      (1) Submarine;

      (2) Watch teams;

      (3) References; and

      (4) Supervision.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) At Sea;

      (2) Watch Dived; and

      (3) Day or Night.

3. Standard. The SMWD shall, IAW references, (T0013) conduct eyes only attacks on two ships no closer than 1200 yards by day and 2000 yards by night.

4. References

A1, A2, A8, A10, A11, A14, A23, A26, A28, B3, B6, B17, B21, B24, B24, C1, D1, D5, D7

5. Task Numbers

T0013

6. Training Limitations

Training will be conducted in SCTT.

PO 004 Conduct Operational Administration

1. Performance. Conduct Operational Administration

2. Conditions

   a. Given:

      (1) Submarine;

      (2) References; and

      (3) Supervision.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) At Sea; and

      (2) Alongside.

3. Standard. The SMWD shall, IAW references, (T0009) conduct operational administration by:

   a. Maintaining reports and returns;

   b. (T0009) Maintaining the operational readiness of an assigned section; and

   c. Coordinating submarine records management.

4. References

A8, A11, A13, A14, A16, A23, A24, A30, B2, B3, B6, D1, D6

5. Task Numbers

T0009

6. Training Limitations. Nil.

PO 005 Assist in the Conduct of Submarine Operations

1. Performance. Assist in the Conduct of Submarine Operations

2. Conditions

   a. Given:

      (1) Submarine;

      (2) References; and

      (3) Supervision.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) At Sea; or

      (2) Alongside.

3. Standard. The SMWD, shall, IAW references, assist in the conduct of submarine operations using the Operational Planning Process (OPP) by:

   a. (T0004, T0007) Providing CHOPS/Sonix input to operation/exercise plans, including Implementing operational message traffic; and

   b. (T0008) Executing Operations as CHOPS/Sonix.

4. References

A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11, A12, A13, A14, A15, A16, A17, A18, A19, A20, A21, A22, A23, A24, A25, A26, A27, A28, A29, A30, B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11, B12, B13, B14, B15, B16, B17, B18, B19, B20, B21, B22, B23, B24, B25, C1, C2, C3, D1, D2, D3, D4, D5, D6, D7, D8

5. Task Numbers

T0004, T0007, T0008

6. Training Limitations

Training will be conducted in the SCTT.

CHAPTER 3: ASSESSMENT OF TRAINEE

**AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE CCA,
COMMANDANT CANADIAN FORCES NAVAL OPERATIONS SCHOOL (CFNOS).**

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and the standard listed must be met to achieve this qualification. PCs will be administered in the form of (practical or written) examinations and the trainee must obtain a pass in all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative tests that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and trainees. Each EO must have at least one EC. Marks for ECs are not calculated in the trainee's final mark.

3. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be done IAW CFITES Vol. 7 – Evaluation

5. Training Development Division shall design all Performance Checks (PCs). All PCs shall be administered or invigilated and scored by a non-involved instructor designated by the senior instructor. Enabling Checks are the responsibility of the Instructional Staff.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional and standards staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol 6 - Conduct of Instruction Programmes and TE SOs or SOPs.

7. Continuous monitoring of candidate progress is required on this course in order to provide the following:

   a. Early warning of trainee difficulties;

   b. Feedback on the effectiveness of training;

   c. Information for the TRB or PRB (if required); and

   d. Information to assist in completion of the CF 377 Course Report.

8. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment SOs or SOPs.

9. PC failures and supplemental PC failures are to be handled IAW Training Establishment SOs or SOPs. Supplemental ECs and PCs shall not be given prior to 24 hours having elapsed since trainee was informed of their EC or PC failure.

TRAINING REVIEW BOARD

10. TRBs shall be convened and conducted IAW with NPTORD 4500-54 - Training Review Process.

COURSE REPORTS

11. Course reports (CF 377) shall be prepared and distributed IAW CFAO 26-12 and NPTORD 4500-51 – Course Reporting and Certificates.

12. Trainee will be graded pass or fail. IAW NPTORD 4500-51 – Course Reporting and Certificates, students will not be ranked.

13. Where necessary for Rank Qualifications, the narrative will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and / or incomplete training must also be included in the course report. Record of trainee counselling is to be maintained by the training establishment.

CHAPTER 4: LESSON SPECIFICATIONS

**AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE CCA,
COMMANDANT CANADIAN FORCES NAVAL OPERATIONS SCHOOL (CFNOS).**

EO 001.01 Interpret the Effects of the Environment on Submarine Sensors

1. Performance. Interpret the effects of the environment on submarine sensors.

2. Conditions

   a. Given:

      (1) Classroom; and

      (2) References.

   b. Denied:

      (1) Assistance;

      (2) Supervision.

   c. Environmental:

      (1) Classroom.

3. Standard. The trainee shall, IAW references, interpret the effects of the environment on submarine sensors by:

   a. Analyzing environmental effects on underwater sensors WRT:

      (1) Military oceanography;

      (2) Ocean bottom topography;

      (3) Chemistry of seawater;

      (4) Physics of sound

      (5) Sound velocity;

      (6) Ocean thermal structure;

      (7) Sound paths;

      (8) Bathy observations;

      (9) Acoustic propless modelling;

      (10) Biological oceanography;

      (11) Regional oceanography;

      (12) Fronts and eddies;

(13) Towed sensors positioning;

(14) Shallow water UWW; and

(15) Submarine towed array.

b. Analyzing the environmental effects on sonar range predictions WRT:

(1) Logarithms and decibels;

(2) The sonar equation; and

(3) Range prediction (ARPS).

4. Time Required

a. EO - 43 x 50 min periods;

b. EC - 2 x 50 minute periods; and

c. PC - 4 x 50 min periods.

5. Instructional Methods

a. Interactive lecture - 39 x 50 minute periods; and

b. Demonstration/practice - 4 x 50 minute periods.

6. Training Aids

a. Acoustic Propagation software; and

b. Bathygraphic thermography.

7. Test Details. This EO will be tested in EC 001.01 and PC 001.

8. References. See para 10.

9. Remarks. Nil.

10. Teaching Points

	Description	Instructional Method	Time	References

TP1	Military Oceanography: Describe the tactical significance of the environment for naval operations and the national and international infrastructure which provides environmental support to naval units: • Rationale for studying oceanography; • USN oceanographic infrastructure; • Canadian Naval Oceanographic Organization; • NATO military oceanographic organization; • Definition of military oceanography; and • Military oceanography training in the CF.	Interactive lecture	3 periods	A15, A22, A24, A27, B3, B4, B5
TP2	Ocean Bottom Topography: Explain the significance of the ocean bottom to ASW and sensor performance WRT: • Ocean dimensions; • Bottom features; • Marine Geology; • Significance of ocean bottom to naval operations.	Interactive lecture	3 periods	A8, A15, A16, A22, A24, A27, B3, B4, B7, B19, B23, D5

TP3	Chemistry of seawater: Describe the basic properties of salt and fresh water and the terminology used to describe them WRT: • Water inventory; • Properties of fresh water; • Salinity; • Density; • Water masses; and • OFA.	Interactive lecture	2 periods	A9, A15, A22, A24, A27
TP4	Physics of sound: Explain the fundamental concepts and terminology associated with Underwater Acoustics: • Use of sound in UWW; • Definition of sound; • Vibrations; • Waves; • Sound waves; • Wave fronts and rays; and • Military applications of sonar.	Interactive lecture	2 periods	A8, A9, A15, A16, A22, A24, A27, B3, B4, B5, B7, B19, B23, D5
TP5	Sound velocity: Explain the significance of a sound velocity profile to ASW, the factors affecting it and how it is measured WRT: • Requirement for knowledge of sound speed; • Factors affecting sound speed; and • Means of obtaining a sound velocity profile at sea.	Interactive lecture	2 periods	A8, A9, A15, A16, A22, A24, A27, B3, B4, B5, B7, B19, B23, D5

TP6	Ocean thermal structure (OTS): Describe the structure of the vertical temperature profile of the ocean, the factors which affect it, and its significance to naval operations WRT: • Significance of OTS to UWW; • 3 layer model of OTS to include “Thermocline Circulation” • Factors affecting the temperature structure in the upper zone; • Variability of OTS; • OTS reporting and forecast products; and • Acoustic implications.	Interactive lecture	2 periods	A9, A15, A16, A22, A24, A27, B3, B4, B5, B7, B19, B23, D5

TP7	Sound Paths:
	
	Describe qualitative prediction of the path(s) which sound rays will take from source to receiver and state the impact this has on tactics and sensor performance:
	
	• Snell's law;
	• Sonic layer depth;
	• Submarine's "best depth to evade detection";
	• Principle sound paths, WRT:
	  ○ Direct path;
	  ○ Bottom bounce;
	  ○ Convergence zone (CZ);
	  ○ Sub surface duct;
	  ○ Surface ducts; and
	  ○ Reliable Acoustic Path (RAP).
	• Multi-path propagation;
	• Ray trace display systems; and
	• Tactical interpretation:
	  ○ Short range propagation;
	  ○ Channel axis;
	  ○ Ping stealing;
	  ○ Afternoon effect.
		Interactive lecture	3 periods	A8, A9, A15, A16, A22, A24, A27, B3, B4, B5, B7, B19, B23, D5

TP8	Bathy Observation:
	Describe the equipment, requirement and procedures for taking a bathy observation and reporting it:
	• Equipment:
	  ○ Bathy recorder;
	  ○ SSXBT/SSXVT; and
	  ○ Malfunctions.
	• When to take a bathy;
	• SVP archiving:
	  ○ What happens to the SVPs when the sub returns to home port.
		Interactive lecture	3 periods	A8, A9, A15, A16, A22, A24, A27, B23, D5

TP 9	Acoustic Prop Loss Modelling: Explain and demo the following methods of prop loss modelling and software: • ARP systems; • Ray path plot; • PC geared systems; and • Deficiencies common to ARPS.	Interactive lecture Demo/performance	2 periods 4 periods	A8, A9, A15, A16, A22, A24, A27, B3, B4, B5, B7, B19, B23, D5
TP10	Biological Oceanography: Describe the impact of Marine Organisms on ASW tactics and sensors: • Impact of marine organisms on UWW; • Sonic animals; • Biological false targets and scatterers; • Bioluminescence; and • Fouling.	Interactive Lecture	2 periods	A8, A9, A15, A16, A22, A24, A27, B3, B4, B5, B7, B19, B23, D5

TP11	Regional Oceanography: Describe the typical oceanographic conditions in Canadian operating areas: • Scotian Shelf/Cold Shelf; • Scotian Slope; • Gulf Stream; • Sargasso Sea; • Labrador Sea/Current; • North East Pacific; • North East Atlantic; • North West Pacific; and • Arctic.	Interactive Lecture	2 periods	A15, A22, A27, B5

TP12	Fronts and Eddies: Explain the Acoustic significance of fronts and eddies and their impact on UWW sensors and tactics:
	• Oceanography:
	  ○ Definition of front;
	  ○ Locations;
	  ○ Features; and
	  ○ Types of fronts.
	• Acoustics:
	  ○ Acoustic definition of front:
	  ○ Towed arrays;
	  ○ Active systems;
	  ○ Pro-submarine implications; and
	  ○ Ship handling/navigation implications.
		Interactive Lecture	2 periods	A8, A9, A15, A16, A22, A24, A27, B3, B4, B5, B7, B19, B23, D5

TP13	Shallow Water UWW: Understand the acoustic problems inherent in conducting UWW in shallow water: • Definition of shallow water; and • UWW problems in shallow water.	Interactive Lecture	2 periods	A8, A9, A15, A16, A22, A24, A27, B3, B4, B5, B7, B19, B23, D5
TP14	Satellite Oceanography: • Ocean remote Sensing; • Canadian REA: and • Ocean Intelligence and Environment Sensing.	Interactive Lecture	2 periods	A15, A22, A24, A27, B3, B4, B5, B7
TP15	Logarithms and decibels: Explain the nature of the decibel systems as it applies to Underwater Acoustics: • Logarithms; • Anti-logarithms; • Definition of decibel; • Uses of decibels; • Combining decibels; and • Significance of decibel notation.	Interactive Lecture	2 periods	A15, A16, A22, A24, A27, B3, B4, B5, B7, B19, B23, D5

TP16	The sonar equation:
	
	Explain the sonar equation, the constituent terms and their use in range prediction:
	
	• Introduction to the sonar equations to include:
	  ○ Use of the sonar equations;
	  ○ Development;
	  ○ Equations;
	  ○ Terms;
	  ○ Limitations; and
	  ○ Applications.
	• Source Level (SL) to include:
	  ○ Definition;
	  ○ Source level for active sonars;
	  ○ Radiated noise; and
	  ○ Measurement and documentation.
	• Noise level (NL) to include:
	  ○ Definition;
	  ○ Self noise;
	  ○ Reverberation;
	  ○ Ambient noise; and
	  ○ Implications.
		Interactive lecture	4 periods	A8, A9, A15, A16, A22, A24, A27, B3, B4, B5, B7, B19, B23, D5

	• Array gain, to include:
	  ○ Definition, including basic concepts involved in Beam Forming;
	  ○ Directivity index; and documentation.
	• Detection Threshold (DT), to include:
	  ○ Definition;
	  ○ Decision problem;
	  ○ Computation; and
	  ○ Documentation.
	• Target Strength (TS), to include:
	  ○ Definition;
	  ○ Factors affecting TS; and
	  ○ Documentation.
	• Transmission Loss (TS), to include:
	  ○ Definition;
	  ○ Spreading;
	  ○ Attenuation;
	  ○ Computation; and
	  ○ Variability.			

	• Conclusion:
	  ○ FOM; and
	  ○ Signal excess
			1 period	A8, A9, A15, A16, A22, A24, A27, B3, B4, B5, B7, B19, B23, D5

ENABLING CHECK – EO 001.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 001.01 will be a Written assessment based on content from EO 001.01.
3. Pass mark will be 80%.
4. Time required: 2 x 50 min periods
5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a Written assessment based on content from EO 001.01.
2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.
3. Do not open the booklet until you are told to do so.
4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.
5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the assessment then come back to the problem question(s).
6. Read each question carefully, and all responses fully, before answering the question.
7. If you have any questions during this assessment, raise your hand.
8. References will/will not be provided.
9. Pass mark is 80%.
10. You will have 2 x 50 min periods to complete this assessment.
11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil.

PART IV – ENABLING CHECK LIST

CONTENT	VALUE
METOC / Satellite Oceanography	5%
MILOC Bathy	5%
Ocean Environment	20%
ARPS	5%
Tactical Employment of Sensors	20%
Sonar Equation / FOM	10%
Active / Passive Range Prediction	10%
Tactical Oceanography	25%

EO 001.02 Apply Principles of Sonar Search Procedures

1. Performance. Apply Principles of Sonar Search Procedures

2. Conditions

   a. Given:

      (1) References; and

      (2) Classroom.

   b. Denied:

      (1) Nil

   c. Environmental:

      (1) Classroom.

3. Standard. The trainee shall, IAW references, apply principles of sonar search procedures by:

   a. Reacting to the threat;

   b. Providing advice to command on the use of sensors; and

   c. Employing a sonar search plan.

4. Time Required

   a. EO - 32 x 50 min periods; and

   b. EC - 3 x 50 min periods.

5. Instructional Methods. Interactive lecture with case study.

6. Training Aids

   a. Handouts; and

   b. Case Study.

7. Test Details. This EO will be assessed in EC 001.02 and PC 001.

8. References. See para 10.

9. Remarks. Nil.

10. Teaching Points

	Description	Method	Time	References

TP1	Reacting to a threat	Interactive lecture/case study	2 periods	A15, A16, A22, A24, A27, B3, B4, B7, B19, B23, D5

TP2	Employing an appropriate sonar search plan:
	• Towed array positioning;
	• NB search plans:
	  ○ Probability of detection;
	  ○ Sweep width;
	  ○ Track spacing;
	  ○ Coverage factors;
	  ○ Frequency of coverage; and
	  ○ Building an Area of Probability (AOP).
	• BB search plans.
		Interactive lecture/case study	2 periods	A15, A16, A22, A24, A27, B3, B4, B7, B19, B23, D5

TP3	Conduct IETM Familiarization	Interactive lecture/case study	2 periods	A15, A16, A22, A24, A27, B3, B4, B7, B19, B23, D5
TP4	Employing Multi-Hull functions	Interactive lecture/case study	2 periods	A15, A16, A22, A24, A27, B3, B4, B7, B19, B23, D5
TP5	Common Broadband (CBB) operation	Interactive lecture/case study	2 periods	A15, A16, A24, B3, B7, D5
TP6	Implement Contact Follower Promotion and Features	Interactive lecture/case study	2 periods	A15, A16, A24, B3, B7, D5
TP7	Passive Narrowband (PNB) operation	Interactive lecture/case study	2 periods	A15, A16, A24, B3, B7, D5
TP8	Determine Classification (CLASS)	Interactive lecture/case study	2 periods	A15, A16, A24, B3, B7, D5
TP9	Conduct Noise Monitoring (NM)	Interactive lecture/case study	2 periods	A15

TP10	Utilize Sonar Logger	Interactive lecture/case study	2 periods	A15, A24
TP11	High Frequency Active (HFA) operation	Interactive lecture/case study	2 periods	A24, B3, B6, B7, B14, B16, B23
TP12	AI and R (Active Intercept) operation	Interactive lecture/case study	2 periods	A24, B3, B6, B7, B14, B16, B23
TP13	Employ On-board Trainer/Test Target Insertion (OBT/TTI)	Interactive lecture/case study	2 periods	A24, B3, B6, B7, B14, B16, B23
TP14	Employ DDCS	Interactive lecture/case study	2 periods	A24, B3, B6, B7, B14, B16, B23
TP15	Responsibilities of Sonar Supervisor (SONSYM)	Interactive lecture/case study	2 periods	A24, B3, B6, B7, B14, B16, B23
TP16	Optimize Acoustic Controls	Interactive lecture/case study	2 periods	A24, B3, B6, B7, B14, B16, B22

ENABLING CHECK – EO 001.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 001.02 will be a Written scenario based test based on content from EO 001.02.

3. Pass mark will be 80%.

4. Time required: 3 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.02 will be a Written scenario based test based on content from EO 001.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is 80%

10. You will have 3 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

CONTENT	VALUE
Reacting to specific threat	10%
Towed Array Positioning	20%
BB Search Plans	20%
NB Search Plans	40%
Active Intercept Operations	10%

PART IV – ENABLING CHECK LIST

Nil.

PERFORMANCE CHECK – PO 001

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. PC 001 will be a Scenario-based Written test based on content from PO 001.

3. Pass mark will be 80%.

4. Time required: 4 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001 will be a Scenario-based Written test based on content from PO 001.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is 80%

10. You will have 4 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

CONTENT	VALUE
Interpret oceanographic data	10%
React to a specific threat	10%
Direct tactical employment of sensors (Including towed array positioning)	30%
Generate/brief a Search Plan	30%
Direct broadband/narrowband operations	20%

PART IV – PERFORMANCE CHECK LIST

Nil.

EO 002.01 Operate Submarine Weapons and Weapon Control Systems

1. Performance. Operate Submarine Weapons and Weapon Control Systems

2. Conditions

   a. Given:

      (1) Classroom;

      (2) SCTT;

      (3) References; and

      (4) Supervision.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) SCTT.

3. Standard. The trainee shall, IAW references, operate submarine weapons and weapon control systems, by:

   a. Conducting pre-firing activities;

   b. Firing the weapon, to include hangfire / misfire drills;

   c. Conducting post-firing activities; and

   d. Conducting deliberate attacks.

4. Time Required

   a. EO - 43 x 50 min periods;

   b. EC - 8 x 50 min periods (1 per trainee); and

   c. PC - 8 x 50 min periods (1 per trainee).

5. Instructional Methods

   a. Interactive Lecture; and

   b. Simulation.

6. Training Aids

   a. Weapons Acoustic Preset Program (WAPP)

7. Test Details

   a. This EO will be tested in EC 002.01 and PC 002.

8. References

   a. See para 10.

9. Remarks. For EC and PC Checklists, further information cannot be included in this QSP as ref A10 (CANSADS) is classified.

10. Teaching Points

	Description	Method	Time	References
TP1	Deliberate attacks: • Closing up Checks	Interactive Lecture/ Demo/ Simulation	7 periods	A1, A8, A10, A23, B21, D5, D8
TP2	Pre-launch: • Entering weapon presets; • Configuring CCS 876; • Tube States; • Allocating Tubes; • Assigning tactics; • Exercising Weapon; and • War Cries	Interactive Lecture/ Demo/ Simulation	7 periods	A1, A8, A10, A23, B21, D5, D8
TP3	Firing the Weapon: • War Cries	Interactive Lecture / Demo/ Simulation	7 periods	A1, A8, A10, A23, B21, D5, D8

TP4	Operating weapon: • Pre-enable Run; • Wire Guidance Commands; • Changing Presets; • Enable to Homing; • Manual Re-Attacks; • Auto Guidance Re-Attacks; • Shut-Down; and • War Cries.	Interactive lecture / Demo/ Simulation	7 periods	A1, A8, A10, A23, B21, D5, D8
TP5	Post firing activities: • Cutting the wire; • Post firing the tube; and • Returning CCS-876 to normal configuration.	Interactive lecture/ Demo/ Simulation	7 periods	A1, A8, A10, A23, B21, D5, D8
TP6	Hangfire / Misfire drills	Interactive Lecture/ Demo/ Simulation	7 periods	A1, A8, A10, A23, B21, D5, D8
TP7	Peace Time Safety Firing Regulations (PTSFR)	Interactive lecture	1 period	A12, B21

ENABLING CHECK – EO 002.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCTT

Arrangements and Procedures

2. EC 002.01 will be a Practical test based on content from EO 002.01.

3. Pass mark is 80% (11/13).

4. Time required: 8 x 50 min periods (1 per trainee).

5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.01 will be a Practical test based on content from EO 002.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will not be provided.

9. Pass mark is 80% (11/13).

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
1. Conduct closing-up checks on WCC IAW A10 to include: (i) General; (ii) Initialization page; (iii) WCD page; (iv) TPD page; (v) Ray trace page; (vi) Receive report from WSC; and (vii) Report to AC.			
2. Conduct pre-launch procedures IAW A10 to include: (i) Deliberate Attack; and (ii) Urgent Attack.			
3. Selecting tactics			
4. Recognize preset limitations			
5. Fire			
6. Control weapon during the pre-enable phase			
7. Control the weapon post-enable phase			
8. Conduct manual re-attack			
9. Shut down			
10. War cries IAW A10			
11. Complete the engagement IAW A10			
12. Secondary mode firing			
13. Misfire or Hangfire drill			

ASSESSORS ADDITIONAL REMARKS:

OVERALL ASSESSMENT:

Are 80% (11/13) of the performance factors SAT?	
Yes	No
PASS	**FAIL**

_____ STUDENTS SIGNATURE

_____ STAFF ASSESSOR

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PERFORMANCE CHECK – PO 002

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCTT

Arrangements and Procedures

2. PC 002 will be a Practical test based on content from PO 002

3. Pass mark is 80% SAT (11/13).

4. Time required: 1 x 50 min period per trainee (8 periods total).

5. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 002 will be a Practical test based on content from PO 002.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will not be provided.

9. Pass mark is 80% SAT (11/13).

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

PART IV – PERFORMANCE CHECK LIST

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
1. Conduct closing-up checks on WCC IAW A10 to include: (i) General; (ii) Initialization page; (iii) WCD page; (iv) TPD Page; (v) Ray trace page; (vi) Receive report from WSC; and (vii) Report to AC.			
2. Conduct pre-launch procedures IAW A10 to include: (i) Deliberate Attack; and (ii) Urgent Attack.			
3. Selecting tactics			
4. Recognize preset limitations			
5. Fire			
6. Control weapon during the pre-enable phase			
7. Control the weapon post-enable phase			
8. Conduct manual re-attack			
9. Shut down			
10. War cries IAW A10 s			
11. Complete the engagement IAW A10			
12. Secondary mode firing			
13. Misfire or Hangfire drill			

ASSESSORS ADDITIONAL REMARKS:

OVERALL ASSESSMENT:

Are 80% (11/13) of the performance factors SAT?	
Yes	No
PASS	**FAIL**

_____ _____

STUDENTS SIGNATURE STAFF ASSESSOR

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EO 003.01 Conduct Eyes Only Two Ship Attacks

1. Performance. Conduct Eyes Only Two Ship Attacks

2. Conditions

   a. Given:

      (1) Classroom;

      (2) SCTT;

      (3) References; and

      (4) Stopwatch.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) SCTT.

3. Standard. The trainee shall, IAW references, conduct eyes only two ship attacks on two ships no closer than 1200 yds by day and 2000 yds by night, to include:

   a. Intermittent all round look;

   b. Look interval calculations;

   c. Go deep range calculations;

   d. Safe depth range calculations;

   e. Range rate calculations;

   f. Look planning; and

   g. Attack Triangle.

4. Time Required

   a. EO - 35 x 50 min periods;

   b. EC - 8 x 50 min periods (1 per trainee); and

   c. PC - 8 x 50 min period PC (1 per trainee).

5. Instructional Methods

a. 7 x 50 min periods Interactive Lectures/Demos; and

b. 28 x 50 min periods Practical.

6. Training Aids

a. Classroom; and

b. SCTT.

7. Test Details

a. EO 003.01 will be tested in EC 003.01 and PC 003

8. References

a. See para 10.

9. Remarks. Nil.

10. Teaching Points

	Description	Method	Time	Reference
TP1	Intermittent all round look: • Philosophy; and • ARL interval calculation.	Interactive lecture	1 period	A1, A2, A8, A10, A11, A14, A23, A26, A28, B3, B6, B17, B21, B24, B24, C1, D1, D5, D7
TP2	Look interval calculations: • Exact method; • Go deep range method; and • Range rate method.	Interactive lecture/ Demo performance	1 period 1 period	A1, A2, A8, A10, A11, A14, A23, A26, A28, B3, B6, B17, B21, B24, B24, C1, D1, D5, D7
TP3	Go deep range calculations	Study assignment	Home assignment	A1, A2, A8, A10, A11, A14, A23, A26, A28, B3, B6, B17, B21, B24, B24, C1, D1, D5, D7
TP4	Safe depth calculations: • Rough weather conditions; and • Ship control considerations.	Interactive lecture/ Demo performance	.5 period/ .5 period	A1, A2, A8, A10, A11, A14, A23, A26, A28, B3, B6, B17, B21, B24, B24, C1, D1, D5, D7

TP5	Range rate calculations	Interactive lecture	1 period	A1, A2, A8, A10, A11, A14, A23, A26, A28, B3, B6, B17, B21, B24, B24, C1, D1, D5, D7
TP6	Look planning: • Stopwatch Management; • Target setup; • Sectors; • Range for me; • Milestone Drill; • Q Drill; • ATB checks; • Half all round looks; • Taking a minute; and • Limiter Procedure.	Interactive lecture/ practical	1 period / 27 periods	A1, A2, A8, A10, A11, A14, A23, A26, A28, B3, B6, B17, B21, B24, B24, C1, D1, D5, D7
TP7	Attack triangle	Interactive Lecture	1 period	A1, A2, A8, A10, A11, A14, A23, A26, A28, B3, B6, B17, B21, B24, B24, C1, D1, D5, D7
TP8	Mental math: • Target range; • Distance off track; • Duration of attack; • Speed required; • Track angle; and • Course to close/open.	Interactive Lecture/ Practical	1 period / in conjunction with TP6	A1, A2, A8, A10, A11, A14, A23, A26, A28, B3, B6, B17, B21, B24, B24, C1, D1, D5, D7

ENABLING CHECK – EO 003.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCTT

Arrangements and Procedures

2. EC 003.01 will be a Practical test based on content from EO 003.01.

3. Pass mark is all critical performance factors SAT with 80% (17/21) of remaining performance factors.

4. Time required: 1 x 50 min period per trainee (8 periods total).

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.01 will be a Practical test based on content from EO 003.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is all critical performance factors SAT with 80% (17/21) of remaining performance factors.

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

*** Indicates critical assessment factors

PART IV – ENABLING CHECK LIST

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
1. Conduct proper ARL at the start of run			
2. Conduct HPL down the bearing of the contacts			
3. Correctly identifies contacts			
4. Conducts target setup on contacts			
5. Correctly calculates DOT			
6. Consistently and correctly calculates LI			
7. *** **Does not go over ARL**			
8. *** **Does not go over the LI of the contacts**			
9. Identifies sector changes			
10. Identifies MARL and update opportunities			
11. Correctly orders MARL			
12. Identifies ATB check opportunities			
13. Correctly applies ATB check criteria			
14. Identifies HARL opportunities			
15. Correctly relaxes from MARL			
16. Mental math speed			
17. Mental math accuracy			
18. Correctly employs stopwatches			
19. Gets away from the periscope where possible			

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
20. Checks bubble, depth, course, speed for every mast exposure			
21. Correct speed for ARL / MARL			
22. Uses proper ranging height			
23. Control Room presence			

***Critical performance factors

ASSESSORS ADDITIONAL REMARKS:

OVERALL ASSESSMENT:

Are all critical performance factors and 80% (17/21) of the remaining performance factors SAT?	
Yes	No
PASS	**FAIL**

_____ STUDENTS SIGNATURE

_____ STAFF ASSESSOR

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PERFORMANCE CHECK – PO 003

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCTT

Arrangements and Procedures

2. PC 003 will be a Practical test based on content from PO 003.

3. Pass mark is all critical performance factors SAT with 80% (17/21) of remaining performance factors.

4. Time required: 1 x 50 min period per trainee (8 periods total).

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 003 will be a Practical test based on content from PO 003

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is all critical performance factors SAT with 80% (17/21) of remaining performance factors.

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

*****Indicates critical performance factor**

PART IV – PERFORMANCE CHECK LIST

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
1. Conduct proper ARL at the start of run			
2. Conduct HPL down the bearing of the contacts			
3. Correctly identifies contacts			
4. Conducts target setup on contacts			
5. Correctly calculates DOT			
6. Consistently and correctly calculates LI			
7. *** **Does not go over ARL**			
8. *** **Does not go over the LI of the contacts**			
9. Identifies sector changes			
10. Identifies MARL and update opportunities			
11. Correctly orders MARL			
12. Identifies ATB check opportunities			
13. Correctly applies ATB check criteria			
14. Identifies HARL opportunities			
15. Correctly relaxes from MARL			
16. Mental math speed			
17. Mental math accuracy			
18. Correctly employs stopwatches			

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
19. Gets away from the periscope where possible			
20. Checks bubble, depth, course, speed for every mast exposure			
21. Correct speed for ARL / MARL			
22. Uses proper ranging height			
23. Control Room presence			

*****Indicates critical performance factor**

ASSESSORS ADDITIONAL REMARKS:

OVERALL ASSESSMENT:

Are all critical performance factors and 80% (17/21) of the remaining performance factors SAT?	
Yes	No
PASS	**FAIL**

_____ STUDENTS SIGNATURE

_____ STAFF ASSESSOR

THIS FORM TO BE RETAINED ON COURSE FILES IAW CFNOSSOs

EO 004.01 Maintain Reports and Returns

1. Performance. Maintain Reports and Returns

2. Conditions

   a. Given:

      (1) Classroom; and

      (2) References.

   b. Denied:

      (1) Assistance.

   c. Environmental:

      (1) Classroom.

3. Standard. The trainee shall, IAW references, maintain reports and returns by:

   a. Ensuring the production and completion of sectional periodic reports and returns; and

   b. (T0009) Maintaining operational readiness of a section:

      (1) Planning a section training plan;

      (2) Implementing a section training plan;

      (3) Section manning / qualification requirements; and

      (4) CRRs.

4. Time Required

   a. EO - 4 x 50 min periods;

   b. EC - 1 x 50 min period; and

   c. PC - 2 x 50 min periods.

5. Instructional Methods

   a. Interactive lecture.

6. Training Aids

   a. Report templates.

7. Test Details

   a. EO 004.01 is tested in EC 004.01 and PC 004.

8. References

   a. See para 10.

9. Remarks. Nil.

10. Teaching Points

	Description	Method	Time	Reference
TP1	Periodic reports and returns	Interactive lecture	1 period	A11, A14, A13, A15, A17, A24, A25, B2, B3,
TP2	Sectional Training responsibilities WRT: • Building a training plan; • Considering OPSKED; • Duties and responsibilities of senior personnel in the Ops Department: - Combat Officer - Sonar Officer - Navigating Officer - Chief of Operations - Senior Sonar Operator (UC1) - Senior Nav Comm (RS) - Senior NCIOP (EW1) • Implementing a training plan.	Interactive lecture	1 period	A11, A14
TP3	Sectional manning requirements WRT: • Watch and station bill; and • SUBSAFE personnel pillar / qualification matrix	Interactive lecture	1 period	A14

TP4	Combat Readiness Requirements (CRR) • Under Water Section CRR's; and • References.	Interactive lecture	1 period	A11

ENABLING CHECK – EO 004.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 004.01 will be a Written test based on content from EO 004.01.

3. Pass mark will be 80%.

4. Time required: 1 x 50 min period.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 004.01 will be a Written test based on content from EO 004.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is 80%.

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

CONTENT	VALUE
1. Identify relevant submarine periodic reports and returns, and indicate the content.	20%
2. Indicate references for combat readiness requirements and describe those relevant to the Under Water Section.	20%
3. Describe the Under Water Section manning requirements IAW CFCD 110(A) (Watch and Station Bill / Qualification Matrix).	20%
4. Describe the duties and responsibilities of senior personnel in the Ops Department: - Combat Officer - Sonar Officer - Navigating Officer - Chief of Operations - Senior Sonar Operator (UC1) - Senior Nav Comm (RS) - Senior NCIOP (EW1)	20%
5. Build a sectional training plan (scenario based).	20%

EO 004.02 Coordinate Submarine Records Management

1. Performance. Coordinate Submarine Records Management

2. Conditions

   a. Given:

      (1) Classroom;

      (2) References; and

      (3) Supervision.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) Classroom.

3. Standard. The trainee shall, IAW references, coordinate submarine records management.

4. Time Required

   a. EO - 3 x 50 min periods; and

   b. EC - 1 x 50 min period.

5. Instructional Methods

   a. Interactive lecture.

6. Training Aids

   a. Nil.

7. Test Details

   a. EO 004.02 will be tested in EC 004.02 and PC 004.

8. References

   a. See para 10.

9. Remarks. Nil.

10. Teaching Points

	Description	Method	Time	Reference

TP1	Records Management WRT: • Submarine record forms; • CASEX records requirements • Logging; • Compiling; • Evidence Gathering; • Packaging; and • Distribution.	Interactive lecture	3 periods	A14, A16, A23, A24, A30, B3, B6, D7

ENABLING CHECK – EO 004.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 004.02 will be a Written test based on content from EO 004.02.

3. Pass mark will be 80%.

4. Time required: 1 x 50 min period.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 004.02 will be a Written test based on content from EO 004.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is 80%

10. You will have 1 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

CONTENT	VALUE
Identify references for Submarine Record Forms (SRF's) and describe all four different SRF's. Fill out a given SRF based on given scenario.	50%
Describe records references and requirements for CASEX's.	30%
Explain the evidence gathering process pertaining to submarine operations.	20%

PERFORMANCE CHECK – PO 004

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. PC 004 will be a scenario-based written test using content from PO 004.

3. Pass mark will be 80%.

4. Time required: 2 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 004 will be a scenario-based written test using content from PO 004.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is 80%

10. You will have 2 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

PART IV – PERFORMANCE CHECK LIST

PERFORMANCE FACTORS	VALUE
Build a sectional training plan for the underwater section based on a given scenario. The training plan should include (but not limited to): - CRR requirements IAW CFCD 102; - Personnel status considerations with regards to qualifications, experience, strengths and weaknesses; - OPSKED timing considerations (planned operations, exercises, DWP, WUPs, etc.); and - Available resources.	50%
Outline a record keeping plan for a given operation/exercise based on a given scenario and generate a narrative type record document. The plan should take into consideration the following: - Records requirements IAW relevant references; - Mechanism to ensure consistency of written language and acronyms between various contributors; - The organization(s) where the record(s) will be sent; and - The classification of the record.	50%

EO 005.01 Provide CHOPS/Sonix Input to Ops/Exercise Plans

1. Performance. Provide CHOPS/Sonix Input to Ops/Exercise Plans

2. Conditions

   a. Given:

      (1) Classroom;

      (2) SCTT;

      (3) References; and

      (4) Supervision.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) SCTT.

3. Standard. The trainee shall, IAW references, (T0004) provide CHOPS/Sonix input to ops/exercise plans by:

   a. Applying Operational Planning Procedures (OPP);

   b. Drafting operational messages;

   c. (T0007) Conducting exercise planning;

   d. Requesting exercise support (RFI/RFS);

   e. (T0007) Planning inshore operations;

   f. (T0007) Planning special operations; and

   g. Planning underwater looks.

4. Time Required

   a. EO - 40 x 50 min periods;

   b. EC - 16 X 50 min periods (2 per trainee); and

   c. PC - 38 x 50 min periods (22 planning and 2 per trainee).

5. Instructional Methods

   a. 16 x 50 min periods Interactive lecture; and

   b. 24 x 50 min periods practical.

6. Training Aids

   a. SDM laptop (1 per trainee); and

   b. Classified laptops (1 per trainee).

7. Test Details. This EO will be tested in EC 005.01 and PC 005.

8. References. See para 10.

9. Remarks. Nil.

10. Teaching Points

	Description	Method	Time	Reference
TP1	Applying OPP:	Interactive lecture Practical	4 periods 3 periods	A19
TP2	Drafting operational messages: • SUBNOTE; • SUBNOTE request; • SUBNOTE change request; • SUBNOTE change; • Diving message; • Surfacing message; • Request for service support (RSS); • Exercise support request; • CASEX message; • Daily SITREP; • MARINTREP; • Significant incident report; • OPSTAT unit; and • RFI/RFS.	Interactive lecture Practical	4 periods 3 periods	A13, A14, A18, B1, B2, B3, B4, B5, B6

TP3	Conducting exercise planning WRT: • Intro to exercises; • Safety precautions, operating restrictions and casualty procedures; • Standard exercises and methods of ordering; • Instructions, definitions, safety rules and procedures applicable to sub ASW exercises / coordinated sub ops (S Series); • Communications; and • Records and analysis.	Interactive lecture	4 periods	A13, A14, A18, B1, B2, B3, B4, B5, B6
TP4	Planning inshore operations WRT: • PERIPHOT; • Sensor Drop; and • Agent RV	Interactive lecture / Demo / Performance	2 periods 16 (2 periods per trainee)	A1,A2, A4, A5, A7, A8, A10, A12, A14, A17, A19, A23, A26, A27, A28, A29, A30, B3, B4, B6, B7, B10, B11, B12, B13, B14, B15, B16, B17, B18, B21, B23, B24, B25, C1, C2, C3, D1, D5, D7
TP5	Planning special operations WRT: • Types; • Methods; • Execution; and • Communications	Interactive lecture	1 period	A1,A2, A4, A5, A7, A8, A10, A12, A14, A17, A19, A23, A26, A27, A28, A29, A30, B3, B4, B6, B7, B10, B11, B12, B13, B14, B15, B16, B17, B18, B21, B23, B24, B25, C1, C2, C3, D1, D5, D7

TP6	Planning underwater looks WRT: • Navigation: ○ Look depth calculation • Sensor employment	Interactive lecture Demo / Performance	1 period 2 periods	A1,A2, A4, A5, A7, A8, A10, A12, A14, A17, A19, A23, A26, A27, A28, A29, A30, B3, B4, B6, B7, B10, B11, B12, B13, B14, B15, B16, B17, B18, B21, B23, B24, B25, C1, C2, C3, D1, D5, D7

ENABLING CHECK – EO 005.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCTT

Arrangements and Procedures

2. EC 005.01 will be a Practical test based on content from EO 005.01.

3. Pass mark will be 80%, in addition to SAT on all critical factors (marked by ***).

4. Time required: 2 x 50 min periods per trainee (16 periods total).

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.01 will be a practical test based on content from EO 005.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark will be 80% in addition to SAT on all critical factors (marked by ***).

10. You will have 2 x 50 min periods to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

*** Indicates critical performance factors

PART IV – ENABLING CHECK LIST

This EC will consist of planning and briefing a CASEX

CASEX PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
Briefing			
Situation • Ephemerals • Weather • Bathy			
Exercise			
Purpose			
OCS			
Participating Units			
Timings			
Areas			
Start Positions			
Priorities			
Safe Depth *** • **Safety Separation Calculations** ***			
Safety Course * **			
Conduct • Phases			
Communications			
Relaxations			
Instructions			
Special Instructions			
Submarine Actions at FINEX			
Tactical Employment of the Environment • Range Predictions • Cutoff Freq • SLD • BLD / BED			

CASEX PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
• Traffic Routes • Bottom Type / Features			
Threats / Responses • Air / Surf / Sub			
Weapons Policy • Tube Status • SSE Status			
Domestic Requirements			
Records Requirements			
Overall Brief • Confident and Concise • Comprehensive • Handling of Questions			

Note: ***Indicates a critical performance factor.

ASSESSORS ADDITIONAL REMARKS:

OVERALL ASSESSMENT:

Are all critical performance factors SAT and 80% of the remaining performance factors SAT?	
Yes	No
PASS	**FAIL**

STUDENTS SIGNATURE

STAFF ASSESSOR

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EO 005.02 Execute Operations as CHOPS/Sonix

1. Performance. Execute Operations as CHOPS/Sonix

2. Conditions

   a. Given:

      (1) SCTT;

      (2) Classroom;

      (3) References; and

      (4) Supervision.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) Classroom; and

      (2) SCTT.

3. Standard. The trainee shall, IAW references, (T0008) execute operations as CHOPS/Sonix by implementing:

   a. ROW ASW tactics;

   b. National/ Allied tactics;

   c. Mine Warfare / Avoidance;

   d. TCMs;

   e. Breaking Contact / Evasion and Escape;

   f. Dived RV;

   g. Opposed Transit;

   h. BINT; and

   i. Delivering Threat Briefs.

4. Time Required

   a. EO - 44 x 50 min periods; and

   b. EC - 16 X 50 min periods EC (2 per trainee).

5. Instructional Methods

a. 31 x 50 min periods interactive lectures/demonstration;

b. 11 x 50 min periods practical; and

c. 2 x 50 min periods presentations.

6. Training Aids

a. References; and

b. SCTT.

7. Test Details. This EO will be assessed in EC 005.02 and PC 005.

8. References. See para 10.

9. Remarks. SME for TP 3, 4, 5 and 9.

10. Teaching Points

	Description	Method	Time	Reference

TP1	Command and Control relationships:
	• Task organization and responsibilities WRT:
	  ○ OSE;
	  ○ OCE
	  ○ OTC; and
	  ○ OCS.
	• Types of Command;
	• Structures of Command (NATO, ALLIED, National Maritime):
	  ○ TACOM;
	  ○ TACON;
	  ○ OPCOM; and
	  ○ OPCON;
	  ▪ CHOP
	• Command during support Operations:
	  ○ Area;
	  ○ Associated;
	  ○ Direct; and
	  ○ Intergrated.
	• Command Structures:
	  ○ Canadian; and
	  ○ NATO.
	• Support:
	  ○ SUBOPAUTH, SMAA, SEAC, SOCA, SEC, SAT
	  ○ WSM/PMI
	  ○ Safety Separation
	  ○ Changing depth through zones
	• Communications:
	  ○ Circuits
	  ○ Call Signs
	  ○ Code Words
	  ○ NUCO
	  ○ Autentication
	  ○ On-Top Procedure
	  ○ ESUS
	  ○ Uncle Joe
	  ○ Hooker Procedure
	  ○ Uniform Procedure
	  ○ Mark-Snap Range
		Interactive Lecture	2 periods	A12, A14, A19, B3, B4, B6

TP2	Air/Sub Co-op: • Air/Sub RV; • Operating Areas; • Types; • Procedures; and • Code words.	Interactive Lecture	2 periods	A12, A14, A19, B3, B4, B6

TP3	ROW ASW Tactics, to include:
	• Primary SSN op areas;
	• Primary diesel op areas;
	• SSBN patrol areas;
	• SSGN patrol areas;
	• SSN/SS tactics, to include:
	  ○ Close support to SSBN/SSGN;
	  ○ Defensive barriers;
	  ○ Anti-SSBN; and
	  ○ Tattletale.
	• Search patterns, to include:
	  ○ SSN;
	  ○ SS;
	  ○ SSGN; and
	  ○ SSBN.
	• Attack phase, to include:
	  ○ Weapons launch;
	  ○ Evasion/counter measures; and
	  ○ Sonar state.
	• Passive sonar tactics;
	• Active sonar tactics;
		Interactive Lecture	10 periods	A12, A14, A19, B3, B4, B6

	• Sea control/sea denial areas, to include:
	  ○ Barrier description:
	  • SSN; and
	  • Diesel (rotating).
	  ○ Forward area submarine searches; and
	  ○ Chokepoint surveillance:
	  • SSN; and
	  • SS.
	• Operating depths (tactical); and
	• Operating speeds (tactical), to include:
	  ○ Transit;
	  ○ Search;
	  ○ Evasion; and
	  ○ Cruising.
	• Air Tactics
	  ○ Platforms
	  ○ Disposition
	  ○ Procedures
	  • Search
	  • Localization
	  • Attack
	• Surface Tactics
	  ○ Platforms
	  ○ Disposition
	  ○ Procedures
	  • Search
	  • Localization
	  • Attack			

TP4	National and Allied Tactics:
	• Surface
	  ○ Classification
	  ○ Attacks
	  ○ Plan Black / Red
	  ○ SAU Procedures
	  ○ Manoeuvring
	  ○ TCMs
	  ○ Green grenade procedure
	• Air
	  ○ Procedures
	  • Search
	  • Localization
	  • Attack
	• Submarine
		Interactive Lecture	4 periods	A12, A14, A19, B3, B4, B6

TP5	Mine Warfare / Avoidance:
	• Types of Mines:
	  ○ Operation;
	  ○ Detection; and
	  ○ Detonation.
	• Mine Transit:
	  ○ Q Routing;
	  ○ Self Protective Measures:
	  ▪ Tactical; and
	  ▪ Material.
	  ○ Quiet States;
	  ○ Routine;
	  ○ Shiphandling; and
	  ○ Damage Control.
		Interactive Lecture	3 periods	A12, A14, A19, B3, B4, B6

TP6	TCMs: • Decoys Caps and Lims; and • Manoeuvring.	Interactive Lecture / Demo Performance	1 period / 1 period	A12, A14, A19, B3, B4, B6

TP7	Breaking Contact / Evasion and Escape: • Fixed Wing Aircraft; • Rotary Wing Aircraft; • Surface Ships; • Submarines; and • Granny's Footsteps	Interactive Lecture / Demo Performance	2 periods / 2 periods	A12, A14, A19, B3, B4, B6
TP8	Dived RV Procedure	Interactive Lecture	1 period	A12, B3
TP9	Basic ROE	Interactive Lecture	2 periods	A1, B22

TP10	Opposed Transit:
	• Initial Planning;
	• Conduct of the snort transit:
	  ○ Snorting Policy; and
	  ○ STD Graph.
	• Additional Planning factors:
	  ○ Rough Weather;
	  ○ Sonar Search, SAC and RTPD;
	  ○ ESM Employment;
	  ○ Periscope Employment;
	  ○ Environment Exploitation;
	  ○ Delousing;
	  ○ Evasion Preplans; and
	  ○ Battery Considerations.
		Interactive Lecture /
		Practical
		
		Presentation
			2 periods /
			
			4 periods /
			
			1 period
				A5, A7, A8, A10, A12, A14, A15, A16, A17, A23, A24, A25, A27, A28, A29, B3, B4, B5, B7, B11, B15, B17, B21, B24, C1, D7

TP11	BINT: • Criteria; • Conduct; • Supporting references; and • Checklists.	Interactive Lecture Demo Performance	1 period / 4 periods	A12, A14, A23, B3, B6, D7
TP12	Threat Briefs	Interactive Lecture Presentation	1 period 1 period	C2

ENABLING CHECK – EO 005.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCTT

Arrangements and Procedures

2. EC 002.02 will be a Practical test based on content from EO 005.02.

3. Pass mark will be 80% in addition to SAT on all critical performance factors (indicated by ***).

4. Time required: 16 x 50 min period (2 per trainee).

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 005.02 will be a Practical test based on content from EO 005.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided

9. Pass mark will be 80% in addition to SAT on all critical factors (marked by ***).

10. You will have 2 x 50 min periods to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

*** Indicates a critical performance factor

PART IV – ENABLING CHECK LIST

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
Conduct COMCHECK • Circuits • Call Signs • Code Words • NUCO • Authentication • On-Top Procedure			
SITREP • Start Time • COMEX • Brief (to outstations) • Time Check			
Reactions • ESUS • Uncle Joe • Hooker Procedures • **Uniform Procedure ***** • Mark Snap Range • Attack			
Safety Separation *** • **Horizontal** • **Vertical** • **Combination** • **MLD Application**			
Changing Depth Through Zones • Changing Depth Safety Signals • EOP Considerations			
Conduct an Attack • Green Grenade Procedures			

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
• Command Solution • UWT Codes • Mark Plots • Annotate Records • Quick Look • ME-YOU			
Situational Awareness • Presence of units			
Records Maintenance			
Overall • Recommendations • Control Room Presence • Situational Awareness			

***Indicates critical performance factors

ASSESSORS ADDITIONAL REMARKS:

OVERALL ASSESSMENT:

Are all critical performance factors SAT and 80% of the remaining performance factors SAT?	
Yes	No
PASS	**FAIL**

STUDENTS SIGNATURE

STAFF ASSESSOR

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PERFORMANCE CHECK – PO 005

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCTT; and

2. Classroom.

Arrangements and Procedures

3. PC 005 will be a Practical test based on content from PO 005

4. Pass mark will be 80% in addition to SAT on all critical performance factors (marked by ***).

5. Time required:

   a) 22 x 50 min periods for planning; and

   b) 2 x 50 minute periods per trainee (16 periods total) in SCTT.

6. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 005 will be a Practical test based on content from PO 005.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark will be 80% in addition to SAT on all critical factors (marked by ***).

10. Time required:

    a) 22 x 50 min periods for planning; and

b) 2 x 50 min periods per trainee in SCTT.

11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

*** Indicates a critical performance factor

PART IV – PERFORMANCE CHECK LIST

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
Preparation			
1. Interpret Messages (i) SUBNOTE (ii) BARNSTORM (iii)CASEX 2) Draft Messages (i) Acknowledgement (ii) SGSA Promulgation 3) e			
Snort Transit Plan • Graph			
Operational Briefing			
Situation • Ephemerals • Weather • Bathy			
Mission			
Command and Control • OPCON • TACOM • TACON			
Units • Friendly • Neutral			

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
• Enemy			
Speed/Time/Distance • SOA • Position in the MHN			
Areas			
Last known positions			
Priorities			
Safe Depth *** • **Safety separation calculations** ***			
Instructions			
Intentions / Plan • Attack Teams / Action Sations			
EOP Considerations			
Communications			
Tactical Employment of the environment • Range Predictions • Cutoff Freq • SLD • BLD / BED • SONAR Search plan • Sonar Stand Off distance • Traffic routes • Bottom type / features			
Threats / Responses • Air / Surf / Sub			
Weapons Policy • Tubes status • Tactics • SSE Status • ROE			

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
Domestic Requirements • Battery considerations • Snorting requirements			
Records requirements • ELINT • COMINT • ACINT • VISINT			

Overall Brief
• Presence (to include:
  ○ Eye Contact;
  ○ Proper enunciation;
  ○ Voice control; and
  ○ Composure under pressure)
• Concise
• Comprehensive
• Handling of questions			

Execution			
Closing-Up Checks • Circuits • Call Signs • Code Words • NUCO • Authentication • Records • Navigation • On-Top Procedure			
SITREP • Brief (to outstations) • Time Check			
Reactions			

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
• Racket Dangerous • Visual Detection of an Aircraft • COI Initial Detection • Recommendations • Manoeuvring			
WSM / PMI • **MLD Application *****			
Conduct of the Event • Event Start • Recommendations • Command Solution • Mark Plots • Annotate Records • Event Stop			
Records maintenance			
Overall • Recommendations • Control room presence • Situational awareness			

ASSESSORS ADDITIONAL REMARKS:

OVERALL ASSESSMENT:

Are all critical performance factors SAT and 80% of the remaining performance factors SAT?

Yes	No
PASS	**FAIL**

_____ _____

STUDENTS SIGNATURE STAFF ASSESSOR

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ANNEX A – MAIN REFERENCES

A1 = CF or other official Canadian government publication
A2 = Second and subsequent references in any category
B = Official Allied Force or Allied government publication
C = Commercial publication
D = Other publications

A1	B-GG-005-004/AF-005 Use of Force in CF Ops Vol. 1 and 2
A2	B-GN-040-000/TS-002-4 MLD Manoeuvring Limitation Diagram
A3	C-09-153-003/TS-000 Explosives Safety Manual Vol 3 - Ships
A4	C23-045-000/AG-004 Damage Control (Sea) Vol 4 – Submarine Damage Control
A5	C-26-509-000/MS-001 Technical Operations Manual Submarine Main Battery type 37KR177T
A6	C-29-537-000/MS-001 Air Purification in Submarines (BR 1326)
A7	C-58-407-000/MS-004 Sea Search
A8	C-69-815-000/MS-001 CCS-876
A9	C-74-370-E00/TA-000 Description and Maintenance Instruction Submarine Launched Pyrotechnics
A10	Canadian Submarine Attack Drills (CANSADS)
A11	CFCD 102 RCN Combat Readiness / Training Requirements
A12	CFCD 106 Vol I, II, III MARTIS
A13	CFCD 107 Naval Communications
A14	CFCD 110 Submarine Operating instructions
A15	CFCD 113 Acoustic Information Guide
A16	CFCD 119 Towed Array User's Guide
A17	CFCD 130 Canadian navigation Manual
A18	CFCD 501 CF Communication Operating Instructions
A19	CFJP 5.0 CF Operational Planning Process
A20	MARCORD G-18 Environmental Management of Shipboard Waste
A21	MARLANT VICTORIA Class EMS
A22	MILOC
A23	NCPM 401 VICTORIA Class Combat Information Organization
A24	NCPM 403 Canadian Submarine SONAR Operating manual
A25	NOSTP 420 Submarine Electronic Warfare Guide
A26	NOSTP 440 Submarine Periscope Photography Handbook

A27	RP-33 Fleet Oceanographic and Acoustic Reference Manual
A28	VICTORIA Class Emergency Operating Procedures (VCEOPs)
A29	VICTORIA Class Standard Operating Procedures (VCSOPs)
A30	VICTORIA Class Submarine Standing Orders (VCSSOs)
B1	APP 7 Joint Brevity Code Words Publication
B2	APP 11 NATO Message Catalogue
B3	ATP 18 Allied Manual of Submarine Operations
B4	ATP 28 Allied Anti-Submarine Warfare Manual
B5	ATP 32 NATO Handbook of Military Oceanographic Information and Services
B6	AXP 1 Allied Submarine and Anti-Submarine Exercise Manual
B7	BQQ 10 Interactive Electronic Technical Manual
B8	BR 1966 (21) 002 One Man Control Console
B9	BR 1966 (27) Buoyant Wire Aerial
B10	BR 1966 (9) 02 Introduction to Power Generation, Propulsion and Control
B11	BR 2075 K Main battery Chloride 880T
B12	BR2800 Echo Sounder Type 778
B13	BR 2805 Sonar Type 2008
B14	BR 2805 Sonar type 2040
B15	BR 45 (4) Admiralty Manual of Navigation
B16	BR 8151 Sonar Type 2040
B17	BR 8409 Periscope System UPHOLDER Class
B18	BR 8551 (8) Echo Sounder Type 780
B19	BR 9072 Sonar Type 2046
B20	Comms System Block Diagram DMSS 8-3 Sketch No. 7003B BR 222 (SM)
B21	FMS1 MK 48 Reference Manual Vol1
B22	MC-362/1 NATO Rules Of Engagement
B23	PRS Operator Manual
B24	SMP 68 Conduct of Diesel Submarine Operations
B25	SMP 95 Fishing Vessel Avoidance
C1	ECPINS Operator Manual
C2	Jane’s Fighting Ships (Current Edition)
C3	Kelvin Hughes Radar Type 1007 User Manual Vol I and II
D1	Captain’s Standing Orders (CSOs)

D2	Command Briefing Cards
D3	Compartment Kill Cards
D4	Sea Search II Operator manual
D5	CCS-876 Operator Manual
D6	Submarine Record Forms
D7	Submarine Tactical Operating Procedures (STOPs)
D8	SUBWOPs

ANNEX B – TASK LIST

TASK NO.	TASK ELEMENT	TRG / NO TRG	PO
T0001	Conduct Submarine Navigation	No Trg	
T0002	Conduct Navigational Related Administration	No Trg	
T0003	Provide Advice to Command on Navigation Related Matters	No Trg	
T0004	**Provide Advice to Command on Warfare Related Matters**	**Trg**	**PO 005**
T0005	**Operate Weapons and Weapon Control Systems**	**Trg**	**PO 002**
T0006	**Provide Advice to Command on the Use of Sensors**	**Trg**	**PO 001**
T0007	**Assist in the Planning of Operations and Exercises**	**Trg**	**PO 005**
T0008	**Assist in the Execution of Operations and Exercises**	**Trg**	**PO 005**
T0009	**Maintain Operational Readiness of Assigned Section**	**Trg**	**PO 004**
T0010	**Manage the Flow of Information**	**Trg**	**PO 001**
T0011	Conduct Narrowband (NB) Threat Motion Analysis	No Trg	
T0012	Supervise Submarine Fire Control System (CCS-876) TMA	No Trg	
T0013	**Conduct Eyes Only Attacks** ~~**Maintain Submarine Safety at Periscope Depth**~~	**Trg**	**PO 003**
T0014	Conduct Manual TMA	No Trg	

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 3-14 Dec 12

Location: CFNOS Halifax NS

Board Members:

SURNAME	INITIALS	RANK	MOSID	UNIT
Bryan	P	LCdr	00207-02	CFNOS Sub Div
Collins	P	LCdr	00207-02	CFNOS Sub Div
Mudge	K	LT(N)	00363	CFNOS Stds Div
Pischedda	J	LT(N)	00207-02	CFNOS Sub Div
McGuire		CPO2	00114	ST (A)
Girard	M	PO1	00324	CFNOS Stds Div
Janssen	J	PO1	00299	DMTE

Board Objective: The aim of the board was to combine the current COTW (AJSL) SS, SOQC (AILV) SS and SMWD (AILT) SS into two new Specialty Specifications (Second Officer of the Watch and Submarine Warfare Director).

Specification Task – No Train Decisions:

The tasks selected for training are detailed at Annex B. Tasks not selected for training are listed below with substantiation:

Task No	Task Statement	Substantiation
T0001	Conduct Submarine Navigation	No Trg – Now to be trained on Submarine Dived Navigation Course, post-SMWD.
T0002	Conduct Navigational Related Administration	No Trg – Now to be trained on Submarine Dived Navigation Course, post-SMWD.
T0003	Provide Advice to Command on Navigation Related Matters	No Trg – Now to be trained on Submarine Dived Navigation Course, post-SMWD.
T0011	Conduct Narrowband (NB) Threat Motion Analysis	No Trg – Previously trained on Submarine 2OOW.

T0012	Supervise Submarine Fire Control System (CCS-876) TMA	No Trg – Previously trained on Submarine 2OOW.
T0014	Conduct Manual TMA	No Trg – Previously trained on Submarine 2OOW.

Discussions / Decisions:

3 Dec 2012

The Chair introduced the board and identified that the aim of the board was to combine the current COTW (AJSL) OSS, SOQC (AILV) OSS and SWD (AILT) OSS into two new OSSs (Second Officer of the Watch and Submarine Warfare Director) and move the navigation tasks to the Surface FNO course as a Submarine PO. Since several members of the board had limited experience with QSP Review Boards; the TDO gave a brief WRT the board process and CFITES role.

The Board started with Task Analysis. They created a scalar using all remaining tasks and reviewed the task lists from the applicable OSSs. During this review, several new tasks were added and/or amended. IAW the new MESIP guidelines all levels and Knowledge statements will be removed. Key points/tasks that were introduced were:

Task statements for the new 2OOW SPQR are as follows:

1) T0001 "Use external communications systems".
2) T0002 "Interpret Submarine sensor data".
3) T0003 "Manage Submarine Fire Control System".
4) T0004 "Conduct periscope watchkeeping".
5) T0005 "Calculate submarine position".
6) T0006 "Conduct dived Emergency Operating Procedures (EOPs) as Ship Control Officer of the Watch (SCOOW)".
7) T0007 "Conduct dived Standard Operating Procedures (SOPs) as Ship Control Officer of the Watch (SCOOW)".
8) T0008 "Maintain Trim".
9) T0009 "Supervise Ship Control".
10) T0010 "Coordinate the Watch".
11) T0011 "Conduct dived operations".
12) T0012 "Employ Manoeuvring Limitation Diagram (MLD)".
13) T0013 "Breakdown operational messages".

Following a group discussion WRT specific SSs, the following changes were proposed:

The Submarine Warfare Director Qualification (AILT) the following tasks were proposed for deletion:

1) T0001 "Conduct Submarine Navigation
2) T0002 "Conduct Navigation related administration"
3) T0003 "Provide advice to command on navigation related matters"

4) T0004 “Provide advice to command warfare related matters”
5) T0006 “Provide advice to command on the use of sensors”
6) T0007 “Assist in the planning of operations and exercises”
7) T0008 “Assist in the execution of operations and exercises”
8) T0009 “Maintain operational readiness of assigned section”

The Submarine Chief of the Watch Qualification (AJSL) the following tasks were proposed for deletion:

1) T0002 “Apply principles of the broadcast”
2) T0003 “Apply principles of the Antenna line ups”
3) T0004 “Communicate using low grade codes”
4) T0005 “Communicate using voice procedures”
5) T0006 “Communicate using the Under Water Telephone (UWT)”
6) T0011 “Conduct Target Motion Analysis (TMA)
7) T0012 “Utilize data link information
8) T0013 “Utilize Global Command Control System- Maritime (GCCS-M) information”
9) T0015 “Prepare Weapon for firing”
10) T0016 “Fire the weapon”
11) T0017 “Control weapon post firing”
12) T0019 “Recognize Periscope threats”
13) T0020 “Conduct Ranging Techniques
14) T0021 “Conduct Target setup (TSU)”
15) T0022 “Conduct Radar Cut Procedures”
16) T0023 “Conduct Warner Look/Take”
17) T0024 “Conduct Look Procedures”
18) T0025 “Calculate All Around Look (ARL)”
19) T0026 “Calculate Go Deep Ranges”
20) T0027 “Conduct Periscope Photography”
21) T0028 “Use Thermal Imaging (TI)”
22) T0029 “Conduct Emergency Go Deep (EGD)”
23) T0030 “Conduct dived navigation”
24) T0032 “Conduct Bottom Contour Fixing
25) T0033 “Create Pool of Errors (POE)”
26) T0035 “Conduct Snap Soundings”
27) T0041 “Advise Command on Engineering implications”
28) T0043 “Supervise helm”
29) T0045 “Direct daily user checks”
30) T0047 “Provide recommendations to command on submarine teactics”
31) T0050 “Interpret Advanced Refractive Effects Prediction System (AREPS)”
32) T0051 “conduct Pre Watch Control Room rounds”

The Submarine Officer Qualification (AILV) the following tasks were proposed for deletion:

1) T0001 “Conduct Contact Evaluation with Submarine Fire Control System (CCS-876)
2) T0002 “Maintain trim”
3) T0003 “Operate Echo Sounder”
4) T0004 “Communicate utilizing external comms”
5) T0005 “Conduct Emergency Operating Procedures (EOPs) as Ship Control Officer of the Watch (SOOCW)”
6) T0006 “Conduct Manual Target Motion Analysis (TMA)”
7) T0007 “Conduct dived navigation”
8) T0008 “Conduct periscope watchkeeping”
9) T0009 “Stand Officer of the Day submarine”
10) T0010 “Direct ship control”
11) T0011 “Interpret Ship’s sensor data”
12) T0012 “Conduct Contact Evaluation”
13) T0013 “Conduct Pre/Post Watch Rounds”
14) T0014 “Evaluate Engineering implications”
15) T0015 “Employ Operational references”
16) T0016 “Direct Duty Watch”
17) T0017 “Direct Duty Watch routines”
18) T0018 “Conduct Duty Watch administration”
19) T0019 “Conduct Submarine Rounds Alongside”
20) T0020 “Coordinate Alongside SOPs”
21) T0021 “Initiate Response to Bomb Threat”
22) T0022 “Initiate Response to Environmental Incident”
23) T0023 “Coordinate Alongside SOPs”
24) T0024 “Direct At Sea Watch Routines”
25) T0025 “Conduct At Sea Watch Administration”
26) T0026 “Conduct Record Keeping”
27) T0027 “Stand as a Control Room Watch Keeping Officer at Sea”
28) T0028 “Navigate with Electronic Navigational Aids”
29) T0029 “Conduct Contact Evaluation with Electronic Navigation Aids”
30) T0030 “Conduct Duty Watch EOP Response”
31) T0031 “Conduct SOPs as SCOOW”

The board agreed on the task statements for the 2 new SSs. DMTE 2-2-2-4 PO1 Janssen approved these changes in principle and the board continued with development as if they were approved. New SPQR paperwork will be submitted in conjunction with the QSPs. Scalar development followed with PO 001 for the SCOOW.

4 Dec 12

Scalar development continued with the 2OOW, and 3 POs were completed. EO 001.01 was started.

5 Dec 12

The board continued with the development of EOs, completed EO 001.01, 02, 03 and PC 001. References will be added at the end of the board.

6 Dec 12

Course development continued with PO 002 and 003 and all associated EOs, ECs and PCs.

7 Dec 12 – The board did not sit this day as there was a CFNOS parade and the Jr Ranks Christmas Dinner.

10 Dec12

The QSPWB shifted to the SMWD QSP and developed PO 001 with its associated EOs, ECs and PC. Some concurrent activity was started to develop a Main Reference List for the 2OOW QSP. PO 002 – PO 005 with their associated EOs, ECs and PCs were developed.

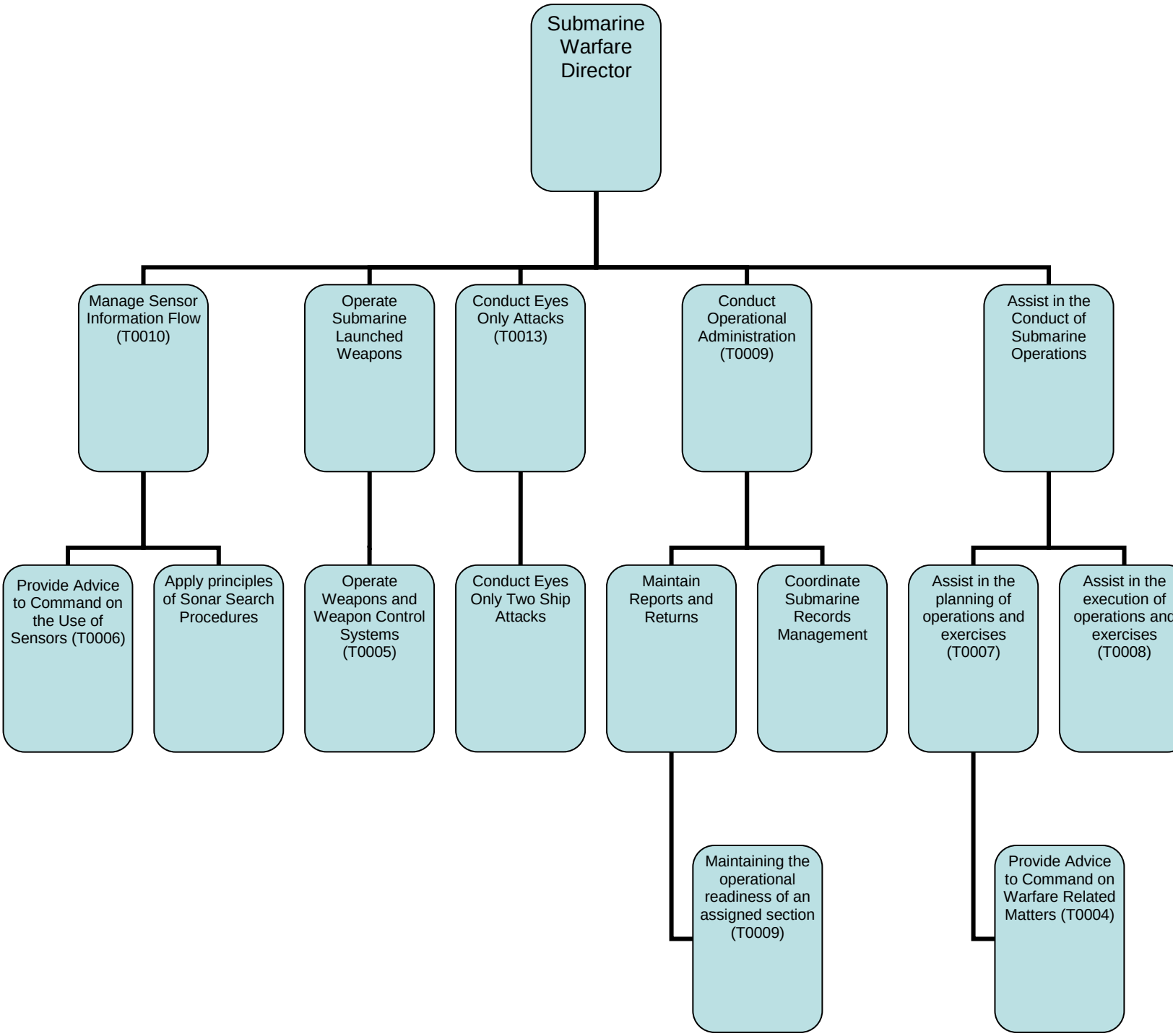
11 Dec 12

Scalars were completed and work continued on the Main Reference List. SPQR paperwork was initiated.

12 Dec 12

Both QSPs were reviewed, SPQR paperwork was reviewed. The main QSPWB was dismissed. Post board activity will continue secretarially.

ANNEX D – QUALIFICATION STANDARD AND PLAN WRITING BOARD SCALAR



Approved 23/11/2016
Version No. 2.5
MITE Crse ID 121009

ANNEX E – ON JOB PERFORMANCE RECORD

NOT ALLOCATED

(Certificate of Qualification)

_____

Name and Initials Rank SN

_____

Date Training Started Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The CCA for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance Objective (PO) to the required standard, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under “Additional Duties” at the end of the OJPR.

PROGESS REVIEW

5. Progress review shall be carried out quarterly by the Divisional Officer and Section Head (NCMs) or Training Officer (Officers),

OJPR Reviewed By (Rank, Position, Name)	Signature	Date

TRAINING COMPLETION CERTIFICATE

Date training completed: _____

_____

"Section Head (NCMs) or HOD (Officers)" Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed training to the standards stated in the Name of QSP except as noted has successfully completed on the job training has successfully completed the Fleet Examination Board. The member is qualified to carry out the duties of job title and is NQual qualified.

_____

Div Officer (NCMs) or School Cmdt or CO (Officers) Date

ON-JOB PERFORMANCE RECORD SHEET

1. The remarks column in this annex shall be used for the following information only:

   a. The unit in which the training and assessment were carried out; and

   b. Any unusual circumstances which preclude compliance with an objective.

ANNEX F – ACRONYMS / ABBREVIATIONS

ACRONYM / ABBREVIATION	TERM/DEFINITION
ACINT	ACOUSTIC INTELLIGENCE
ACU	AIRCRAFT CONTROL UNIT
AIS	AUTOMATIC INFORMATION SYSTEM
AJAG	ASSISTANT JUDGE ADVOCATE GENERAL
ALRS	ADMIRALTY LIST OF RADIO SIGNALS
AML	ADVANCED MILITARY LAYER
AOP	AREA OF PROBABILITY
ARPA	AUTOMATIC RADAR PLOTTING AID
ARPS	ACOUSTIC RANGE PREDICTION SOFTWARE
ASW	ANTI SUBMARINE WARFARE
ASWFA	ANTI SUBMARINE WARFARE FREE AREA
ATB	ANGLE ON THE BOW
AWNIS	ALLIED WORLDWIDE NAVIGATIONAL INFORMATION SYSTEM (NATO)
BA	BRITISH ADMIRALTY
BB	BROADBAND
BED	BEST EVASION DEPTH
BINT	BASIC INTELLIGENCE GATHERING
BITE	BUILT-IN TEST EQUIPMENT
BLD	BEST LISTENING DEPTH
CANSADS	CANADIAN SUBMARINE ATTACK DRILLS
CATAS	CRITICAL ANGLE TOWED ARRAY SYSTEM
CBTO	COMBAT OFFICER
CCA	CURRICULUM CONTROL AUTHORITY
CCG	CANADIAN COAST GUARD
CCISO	COMMAND AND CONTROL INFORMATION SYSTEMS OFFICER
CFMWC	CANADIAN FORCES MARITIME WARFARE CENTER
CFNES	CANADIAN FORCES NAVAL ENGINEERING SCHOOL
CFNOS	CANADIAN FORCES NAVAL OPERATIONS SCHOOL
CHOP	CHANGE OF OPERATIONAL CONTROL

CHS	CANADIAN HYDROGRAPHIC SERVICE
CNPTG	COMMANDER NAVAL PERSONNEL & TRAINING GROUP
COMINT	COMMUNICATION INTELLIGENCE
COSPAR	COMITEE ON SPACE RESEARCH (SARSAT)
CQD	CLOSE QUARTER DRILL
CR	CONTROL ROOM
CRR	COMBAT READINESS REQUIREMENTS
CTIXSJ	INFORMATION EXCHANGE SYSTEM
CTO	COURSE TRAINING OFFICER
DEMON	DEMODULATION
DGEO OPS	DIRECTORATE OF GEOGRAPHIC OPS
DGPS	DIFFERENTIAL GPS
DIINS	DIGITAL INERTIAL NAVIGATION SYSTEM
DMETOC	DIRECTOR METEROLOGICAL AND OCEANOGRAPHIC CENTER
D NAV P&T	DIRECTOR NAVAL PERSONNEL & TRAINING
DNC	DISTANCE TO NEW COURSE
DOT	DISTANCE OFF TRACK
DR	DEAD RECKONING
DT	DETECTION THRESHOLD
EC	ENABLING CHECK
ECB	ELECTRONIC COMMUNICATION BUOY
ECDIS	ELECTRONIC DISPLAY AND INTEGRATION SYSTEM
EEZ	EXCLUSIVE ECONOMIC ZONE
EFA	ELECTRONIC FIXING AID
EFZ	EXCLUSIVE FISHING ZONE
EGD	EMERGENCY GO-DEEP
ELINT	ELECTRONIC INTELLIGENCE
ENC	ELECTRONIC NAVIGATION CHART
EO	ENABLING OBJECTIVE
EOP	EMERGENCY OPERATING PROCEDURES
EP	ESTIMATED POSITION
EPIRB	EMERGENCY POSITION INDICATING RABIO BEACON

EPM	ELECTRONIC PROTECTIVE MEASURE
ETA	ESTIMATED TIME OF ARRIVAL
ETD	ESTIMATED TIME OF DEPARTURE
EW	ELECTROIC WARFARE
FOM	FIGURE OF MERIT
FOTC	FORCE OVER THE HORIZON TRACK COORDINATOR
FSK	FREQUENCY SHIFT KEYING
FTC-STC	FREQUENCY TIME CONSTANT / SENSITIVITY TIME CONTROL
GM	GREENWICH MERIDIAN
GMDSS	GLOBAL MARITIME DISRTRESS
GPS	GLOBAL POSITIONING SYSTEM
HA	HOME ASSIGNMENT
HBR	HIGH BEARING RATE
HF	HIGH FREQUENCY
HIT	HIGH INTEREST TRACK
HSA	HORIZONTAL SEXTANT ANGLE
HYDROPAC	NAVIGATION WARNING MESSAGE PACIFIC
IALA	INTERNATIONAL ASSOCIATION OF LIGHTHOUSE AUTHORITIES
IAW	IN ACCORDANCE WITH
IMO/IHO	INTERNATIONAL MARITIME ORGANIZATION
INS	INTERNATIONAL NAVIGATION SYSTEM
JTAA	JOINT ACTION AREA
KH	KELVIN HUGHES
LDL	LIMITING DANGER LINE
LF	LOW FREQUENCY
LOF	LLOYDS OPEN FORM
LOGREQ	LOGISTICAL REQUIREMENT MESSAGE
LOP	LOCAL OPERATIONS PLOT
LOPO	LOCAL OPERATIONS PLOT OFFICER
LOS	LINE OF SOUND
METOC	METEROLOGICAL AND OCEANOGRAPHIC CENTER

MHN	MOVING HAVEN
MLD	MANOEUVERING LIMITATION DIAGRAM
MRL	MARITIME REAR LINK
MSK	MULTIPLE SHIFT KEYING
MTI	MOVING TARGET INDICATOR
NABS	NAVIGATION BRIDGE SIMULATOR
NATO	NORTH AMERICAN TREATY ORGANIZATION
NAVAREAS	NAVIGATIONAL AREAS
NAVPIM	NAVIGATIONAL PLANNED INTENDED MOVEMENT MESSAGE
NB	NARROWBAND
NCIOP	NAVAL COMBAT INFORMATION OPERATOR
NET	NAVAL ELECTRONIC TECHNICIAN
NLT	NO LESS THAN
NMEA	NATIONAL MARINE ELECTRONICS ASSOCIATION
NMT	NO MORE THAN
NOI / TSEA	NOTICE OF INTENT / TEMPORARY SUBMARINE EXERCISE AREA
NOTACK	WATERSPCE AMANAGEMENT MESSAGE
NPTT	NAVAL PART TASK TRAINEE
NTI	NAVAL TRAINING INSTRUCTOR
NUCO	NATO UNIFORM CODE FOR OPERATIONS
OCE	OFFICER CONDUCTING THE EXERCISE
OCS	OFFICER CONDUCTING THE SERIAL
OFA	OCEAN FEATURES ANALYSIS
OGD	OTHER GOVERNMENT DEPARTMENT
OOW	OFFICER OF THE WATCH
OPCOM	OPERATIONAL COMMAND
OPCON	OPERATIONAL CONTROL
OPSKED	OPERATIONAL SCHEDULE
ORO PIP	OPERATIONS OFFICER PRE INSTRUCTIONAL PACKAGE
OSA	OWN SPEED ACROSS LINE OF SOUND

OSE	OFFICER SCHEDULING THE EXERCISE
OTC	OFFICER IN TACTICAL COMMAND
OTCIXS	INFORMATION EXCHANGE SYSTEM
OTS	OCEAN THERMAL STRUCTURE
OTSR	OPTIMUM TRACK SHIP ROUTING
PC	PERFORMANCE CHECK
PERIPHOT	PERISCOPE PHOTOGRAPHY OPERATION
PMI	PREVENTION OF MUTUAL INTERFERENCE
PO	PERFORMANCE OBJECTIVE
POE	POOL OF ERRORS
PSC	PLOT SENSOR COORDINATOR
PSEA	PERMANENT SUBMARINE EXERCISE AREA
QSP	QUALIFICATION STANDARD & PLAN
RAP	RELIABLE ACOUSTIC PATH
RBROTSARC	RANGE/ BEARING RATE/ OSA/ TSA/ SPEED/ ATB/ RECIPROCAL/ COURSE
RCMP	ROYAL CANADIAN MOUNTED POLICE
RCU/RCVU	RECEIVER CONTROL (VOICE) UNIT
REA	RAPID ENVIRONMENTAL ASSESSMENT
RFI	REQUEST FOR INFORMATION
RFS	REQUEST FOR SUPPORT
ROE	RULES OF ENGAGEMENT
ROW	REST OF WORLD
RSS/ESR	REQUEST FOR SERVICE SUPPORT / EXERCISE SUPPORT REQUEST
RTPD	RETURN TO PERISCOPE DEPTH
SAA	SUBMARINE ACTION AREA
SAC	STERN ARC CLEARANCE
SAMO	SPEED/ADD/MEAN/OVERALL
SAR	SEARCH AND RESCUE
SARSAT	SEARCH AND RESCUE SATELLITE
SART	SEARCH AND RESCUE TRANSPONDER
SAT	SATISFACTORY

SATCOM	SATELLITE COMMUNICATIONS
SDA	SUBMARINE DANGER AREA
SDM	SHINADS DUAL MILITARY
SEAC	SUBMARINE EXERCISE AREA COORDINATOR
SEC	SUBMARINE ELEMENT COORDINATOR
SEOSS	SHIPBOARD ELECTRO-OPTIC SURVEILLANCE SYSTEM
SEPIRB	SUBMARINE ELETRONIC POSITIONAL INDICATING RADIO BEACON
SGSA	SUBMARINE GENERATED SEARCH AREA
SHINADS	SHIPBOARD INTEGRATED NAVIGATION DISPLAY SYSTEM
SHM	SHIPS HEAD MARKER
SL	SOURCE LEVEL
SLD	SONIC LAYER DEPTH
SMAA	SUBMARINE MOVEMENT ADVISORY AUTHORITY
SMWD	SUBMARINE WARFARE DIRECTOR
SOA	SPEED OF ADVANCE
SOCA	SUBMARINE OPERATION COORDINATING AUTHORITY
SOLAS	SAFETY OF LIFE AT SEA
SOP	STANDARD OPERATING PROCEDURE
SPA	SUBMARINE PATROL AREA
SS	SPECIALTY SPECIFICATION
SSE	SUBMARINE SIGNAL EJECTOR
SSL	SUBMARINE SAFETY LANE
SSO	SHIP'S / SCHOOL STANDING ORDERS
STD	SPEED TIME DISTANCE
SUBAUTH	SUBMARINE AUTHORITY
SUBNOTE	SUBMARINE NOTICE
TA	TOWED ARRAY
TACO	TACTICAL OFFICER
TACOM	TACTICAL COMMAND
TACON	TACTICAL CONTROL

TBP	TIME BEARING PLOT
TDS	TACTICAL DATA SYSTEM
TGSA	TASK GROUP SEARCH AREA
TL	TRANSMISSION LOSS
TMA	TARGET MOTION ANALYSIS
TOAST	TACTICAL OCEANOGRAPHIC AND ATMOSPHERIC SUPPORT TECHNOLOGIES
TSA	TARGET SPEED ACROSS LINE OF SOUND
TTY	TELETYPE
UHF	ULTRA HIGH FREQUENCY
UWT	UNDERWATER TELEPHONE
UWW	UNDER WATER WARFARE
VCS	VICTORIA CLASS SUBMARINE
VISINT	VISUAL INTELLIGENCE
VDS	VERIABLE DEPTH SONAR
VHF	VERY HIGH FREQUENCY
VSA	VERTICAL SEXTANT ANGLE
VSCTT	VICTORIA CLASS CUBMARINE COMMAND TEAM TRAINER
VTMS	VESSEL TRAFFIC MANAGEMENT SYSTEM
WSM	WATERSPACE MANAGEMENT
WZ	Prefix for Navigation Warning

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

PO/EO	PO / EO TITLE	EO (x50 min periods)	EC* (x50 min periods)	PC* (x50 min periods)	Total (x50 min periods)
EO 001.01	Interpret the Effects of the Environment on Submarine Sensors	43	2		
EO 001.02	Apply Principles of Sonar Search Procedures	32	3		
PO 001	**Manage Sensor Information Flow**	**75**	**5**	**4**	**84**
EO 002.01	Operate Submarine Weapons and Weapon Control Systems	43	8 (1 per trainee)		
PO 002	**Operate Submarine Launched Weapons**	**43**	**8**	**8 (1 per trainee)**	**59**
EO 003.01	Conduct Eyes Only Attacks	**35**	8 (1 per trainee)		
PO 003	**Conduct Eyes Only Attacks**	**35**	**8**	**8 (1 per trainee)**	**51**
EO 004.01	Maintain Reports and Returns	4	1		
EO 004.02	Coordinate Submarine Records Management	3	1		
PO 004	**Conduct Operational Administration**	**7**	**2**	**2**	**11**
EO 005.01	Provide CHOPS/Sonix input to ops/exercise plans	40	16 (2 per trainee)		
EO 005.02	Execute Operations as CHOPS/Sonix	44	16 (2 per trainee)		
PO 005	**Assist in the Conduct of Submarine Operations**	**84**	**32**	**38 (22 planning+2 per trainee in SCTT)**	**154**
SMWD COURSE TOTAL		**244**	**55**	**60**	**359**

ANNEX H – ADDITIONAL TRAINING RESOURCES

In accordance with Chapter 1 paragraph 3 add any extra tables or paragraphs used for Resources in this annex. (**Not** to be deleted.)

ANNEX I – COSTING INFORMATION

Include information about cost in whichever format is applicable.

This annex is optional (**Not** to be deleted).

Estimates only, based on CF Cost Factors Manual FY 2012/13 verify that this link is the most updated version

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: TDL 300		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	14	$1,000.0		$14,000	$14,000	$70,000.00
R&Q	14	$36.0	20	$10,080	$10,080	$50,400.00
Incidentals	14	$17.5	20	$4,900	$4,900	$24,500.00
Subtotal				**$28,980**	**$28,980**	**$144,900.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	16	$50.0	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	16	$1,000.0	1	$16,000	$0	$16,000.00
IT Equip (Proj, Smart board etc)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.0	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc)	15	$30.0	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2	2	$223.0	20	$8,920	$8,920	$44,600.00
Incremental Civilian Instructor		XXX				
		subtotal		**$8,920**	**$8,920**	**$44,600.00**
	Total Estimated Costs			**$66,165**	**$39,165**	**$222,825**

ANNEX J – COURSE HISTORY

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update / modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Detail as to the nature and reasons for amendments is to be provided here.

Form found in NPTORD 4500-22 Annex B

NPTORD 4500-22
Annex B_QSP Amend

The course was first run in 2005 with five trainees and then again in 2008 with three trainees. Given the small number of trainees and the fact that no pilot course was done and the instructors' notes were lost there is no recorded course history.
The QSPWB convened in January 2009 with essentially a blank sheet from which to start constructing the course.

Jun 2015

The SMWD was once again reviewed to finalize decisions rendered in 2012. With the impending creation of the Submarine Dived Nav course (to be trained post-FNO/SMWD), and the training of TMA tasks on the 2OOW, all Navigation and TMA-related tasks (T0001-T0003 and T0011, T0013-14) were amended to be NO TRAIN within the SMWD QSP.

28 Nov 16- QSP Approved by CCSF

CR-Submarine
Warfare Director (AIL

ANNEX K – ASSESSMENT PLAN

EC / PC	Scope Covered	How	Time (x50 min Periods)	Pass / Fail Criteria	Resources required	Limitations
EC 001.01	EO 001.01	Written	2	80%	Classroom	Nil
EC 001.02	EO 001.02	Written	3	80%	Classroom	Nil
PC 001	PO 001	Written	4	80%	Classroom	Nil
EC 002.01	EO 002.01	Practical	8 (1 per trainee)	80% (EC list)	SCTT	SCTT availability, restricted access area
PC 002	PO 002	Practical	8 (1 per trainee)	80% (PC list)	SCTT	SCTT availability, restricted access area
EC 003.01	EO 003.01	Practical	8 (1 per trainee)	80% (EC list)	SCTT	SCTT availability, restricted access area
PC 003	PO 003	Practical	8 (1 per trainee)	80% (PC list)	SCTT	SCTT availability, restricted access area
EC 004.01	EO 004.01	Written	1	80%	Classroom	Nil
EC 004.02	EO 004.02	Written	1	80%	Classroom	Nil
PC 004	PO 004	Written	2	80%	Classroom	Nil
EC 005.01	EO 005.01	Practical	16 (2 per trainee)	80% (EC list)	SCTT	SCTT availability, restricted access area
EC 005.02	EO 005.02	Practical	16 (2 per trainee)	80% (EC list	SCTT	SCTT availability, restricted access area
PC 005	PO 005	Practical	38 (22 planning + 2 per trainee)	80% (PC list)	SCTT	SCTT availability, restricted access area